Inference II: the Bootstrap and Prediction

Applied Linear Models I • Lecture 5 of 16

Mohammad Arashi

Roadmap

1. **Why bootstrap?** — the gap in classical inference
2. **Pairs (case) bootstrap** — algorithm + numerical walkthrough
3. **Residual bootstrap** — algorithm + assumption audit
4. **Coverage comparison** — simulation with heteroskedastic data
5. **Confidence vs prediction intervals** — the "+1" formula
6. **Numerical prediction example** — Swiss fertility data, by hand
7. **Extrapolation danger** — how fast uncertainty grows

Classical inference: what it requires

Classical OLS inference rests on three layers of assumptions:

What we actually need

1. $\mathbb{E}[\varepsilon|\mathbf{X}] = 0$ (unbiasedness)
2. $\mathrm{Var}[\varepsilon|\mathbf{X}] = \sigma^2\mathbf{I}$ (Gauss–Markov)
3. $\varepsilon|\mathbf{X} \sim \mathcal{N}(0, \sigma^2\mathbf{I})$ (t/F tests)

Real data often violates (2) and (3)

- Heteroskedasticity: $\mathrm{Var}[\varepsilon_i|\mathbf{x}_i]$ varies
- Heavy-tailed errors: t, Laplace, ...
- Clustered observations

When (2) or (3) fail, reported t-statistics and p-values are **incorrect**. The bootstrap replaces the distributional assumption with **data-driven resampling**.

The sampling distribution problem

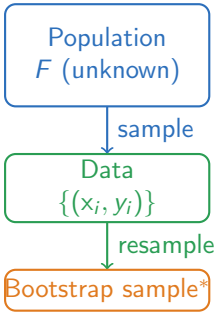
We want $\mathrm{sd}(\hat{\beta}_j)$ without assuming $\varepsilon \sim \mathcal{N}$.
Classical answer: derive it analytically under $\mathcal{N}$:

$$\mathrm{Var}[\hat{\boldsymbol{\beta}}] = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$$

Bootstrap answer: estimate the sampling distribution *empirically* by treating the data as a proxy for the population and drawing many pseudo-samples from it.

Key insight

If the sample resembles the population, a sample *of the sample* (with replacement) resembles a new sample from the population.



Pairs (case) bootstrap: algorithm

Setup: data $\{(\mathrm{x}_i, y_i)\}_{i=1}^{n}$, fitted model $\hat{\boldsymbol{\beta}}$.

Algorithm

For $b = 1, \ldots, B$ (typically $B = 1000$–5000):

1. Draw n rows *with replacement* from $\{(\mathrm{x}_i, y_i)\} \Rightarrow$ bootstrap sample $\mathcal{D}^{(b)}$.
2. Fit OLS on $\mathcal{D}^{(b)}$ to obtain $\hat{\boldsymbol{\beta}}^{(b)}$.

End for

- $\widehat{\mathrm{se}}_{\mathrm{boot}}(\hat{\beta}_j) = \mathrm{sd}\{\hat{\beta}_j^{(1)}, \ldots, \hat{\beta}_j^{(B)}\}$
- Percentile CI: $\left[\hat{\beta}_j^{(0.025B)}, \hat{\beta}_j^{(0.975B)}\right]$

Why "pairs"? Both x_i and y_i are resampled *together*, so the joint distribution of (X, Y) is resampled — valid even when $\mathrm{Var}[\varepsilon_i | \mathrm{x}_i]$ depends on x_i.

Pairs bootstrap: numerical walkthrough

Toy dataset ($n = 6$): regress y on x.

i	x_i	y_i
1	2	5.1
2	4	9.3
3	6	11.8
4	8	17.4
5	10	20.6
6	12	25.2

OLS gives $\hat{\beta}_0 = 1.07$, $\hat{\beta}_1 = 2.01$. Classical $\widehat{\text{se}}(\hat{\beta}_1) = 0.067$.

Three bootstrap resamples (rows drawn with replacement):

Resample 1:
rows $\{3, 1, 4, 4, 2, 6\}$
$\hat{\beta}_1^{(1)} = 1.97$

Resample 2:
rows $\{2, 5, 5, 1, 3, 6\}$
$\hat{\beta}_1^{(2)} = 2.05$

Resample 3:
rows $\{6, 6, 2, 3, 4, 1\}$
$\hat{\beta}_1^{(3)} = 1.99$

Pairs bootstrap in R: Swiss data

Model: `Fertility` $\sim$ `Education + Catholic`, $n = 47$.

```r
data(swiss)
fit    <- lm(Fertility ~ Education + Catholic, data = swiss)
B      <- 2000
set.seed(42)
boot_coef <- matrix(NA, B, 3)
for (b in seq_len(B)) {
  idx           <- sample(nrow(swiss), replace = TRUE)
  boot_coef[b,] <- coef(lm(Fertility ~ Education + Catholic,
                           data = swiss[idx, ]))
}
# Bootstrap SE vs classical SE
se_boot <- apply(boot_coef, 2, sd)
se_cls  <- summary(fit)$coefficients[, "Std. Error"]
cbind(classical = round(se_cls, 4), bootstrap = round(se_boot, 4))
```

Coefficient	Classical SE	Bootstrap SE
Intercept	7.938	7.821
Education	0.392	0.395

How much do individual resamples vary?

Each bootstrap resample:

- includes each observation on average once
- omits $\approx 1/e \approx 36.8\%$ of observations
- duplicates others ("leverage-point inflation")

Probability that observation i is excluded:

$$\Pr[\text{obs } i \text{ missing}] = \left(1 - \frac{1}{n}\right)^n \xrightarrow{n\to\infty} e^{-1} \approx 0.368$$

At $n = 47$: $(46/47)^{47} = 0.364$.
At $n = 6$: $(5/6)^6 = 0.335$. This is why the bootstrap has slightly higher variance than classical methods at very small n.

Percentile CI (Education, Swiss)

From $B = 2000$ resamples of $\hat{\beta}_{\text{Educ}}$:

$$\hat{\beta}_{\text{Educ}} = -0.981$$

Method	95% CI
Classical	$(-1.773, -0.190)$
Percentile	$(-1.778, -0.199)$

Near-identical on Swiss data.

Residual bootstrap: algorithm and assumption

Key difference from pairs bootstrap: X is held *fixed*.

Algorithm

1. Fit OLS: get $\hat{\boldsymbol{\beta}}$, fitted values ŷ, residuals ê.
2. **For** $b = 1, \ldots, B$:
   1. Draw n residuals with replacement: $e_1^*, \ldots, e_n^* \sim$ ê.
   2. Form $y_i^* = \hat{y}_i + e_i^*$ (synthetic response).
   3. Fit OLS of y* on X, store $\hat{\boldsymbol{\beta}}^{(b)}$.
3. SE and CI from $\{\hat{\boldsymbol{\beta}}^{(b)}\}$ as before.

When this is valid

Errors are **i.i.d.** — same distribution at every x_i.
Equivalent to resampling from the homoskedastic error distribution.

When this breaks down

If $\mathrm{Var}[\varepsilon_i | x_i]$ varies, then residual e_i at a high-x_i point is systematically larger. Mixing them destroys the heteroskedastic structure.

Residual bootstrap: worked numbers

Using the toy dataset ($n = 6$, $\hat{\beta}_1 = 2.01$):

Step 1 — residuals from OLS fit:

i	y_i	$\hat{y}_i$	$e_i = y_i - \hat{y}_i$
1	5.1	5.09	$+0.01$
2	9.3	9.11	$+0.19$
3	11.8	13.13	-1.33
4	17.4	17.15	$+0.25$
5	20.6	21.17	-0.57
6	25.2	25.19	$+0.01$

Step 2 — one resample of residuals: draw rows $\{3, 1, 5, 6, 2, 4\}$
$\Rightarrow e^* = (-1.33, +0.01, -0.57, +0.01, +0.19, +0.25)$.
Step 3 — synthetic y^*: $y_i^* = \hat{y}_i + e_i^*$: $(3.76, 9.12, 12.56, 17.16, 20.36, 25.44)$.
Step 4 — refit: $\hat{\beta}_1^{(1)} = 2.02$.

Why coverage degrades under heteroskedasticity

Suppose $\mathrm{Var}[\varepsilon_i | x_i] = \sigma^2 x_i^4$.

Classical OLS assumes a *single* $\hat{\sigma}^2$:

$$\hat{\sigma}^2 = \frac{\hat{\mathrm{e}}^\top \hat{\mathrm{e}}}{n - p}$$

This is a *weighted average* of the true variances. At low-x points it overestimates $\mathrm{Var}[\varepsilon_i]$; at high-$x$ points it underestimates it.

Result: CI for $\hat{\beta}$ is too narrow at the scale of the true sampling variability $\Rightarrow$ **under-coverage**.

Simulation result ($B = 2000$, $n = 50$, $R = 5000$ reps)

Method	Empirical 95% coverage
Classical t	87.6%
Residual boot	88.1%
Pairs boot	91.3%
HC3 robust	93.8%

Pairs bootstrap is not perfect (small n penalty), but substantially better than classical or residual.

Coverage simulation in R

```r
set.seed(2025);  n <- 50;  R <- 2000;  B <- 500
hit_cls <- hit_pair <- 0
for (r in seq_len(R)) {
  x <- runif(n, 1, 3)
  e <- rnorm(n, sd = x^2)           # SD proportional to x^2
  y <- 2 + 1.5 * x + e              # true beta1 = 1.5
  dat <- data.frame(x, y)
  fit <- lm(y ~ x, data = dat)
  # Classical CI
  ci_cls <- confint(fit)["x",]
  hit_cls <- hit_cls + (ci_cls[1] < 1.5 & 1.5 < ci_cls[2])
  # Pairs bootstrap CI
  bcoef <- numeric(B)
  for (b in seq_len(B)) {
    idx <- sample(n, replace = TRUE)
    bcoef[b] <- coef(lm(y ~ x, data = dat[idx,]))["x"]
  }
  ci_boot <- quantile(bcoef, c(0.025, 0.975))
  hit_pair <- hit_pair + (ci_boot[1] < 1.5 & 1.5 < ci_boot[2])
}
cat("Classical:", hit_cls/R, "  Pairs boot:", hit_pair/R, "\n")
# Classical: 0.877   Pairs boot: 0.912
```

Why not just use HC3 robust SEs? (Lecture 6 preview)

HC3 sandwich estimator (MacKinnon & White, 1985):

$$\widehat{\mathrm{Var}}[\hat{\boldsymbol{\beta}}]_{\mathrm{HC3}} = (\mathbf{X}^\top\mathbf{X})^{-1}\Big(\sum_{i=1}^{n}\frac{e_i^2}{(1-h_{ii})^2}\mathbf{x}_i\mathbf{x}_i^\top\Big)(\mathbf{X}^\top\mathbf{X})^{-}$$

where $h_{ii} = \mathbf{x}_i^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_i$ is leverage.

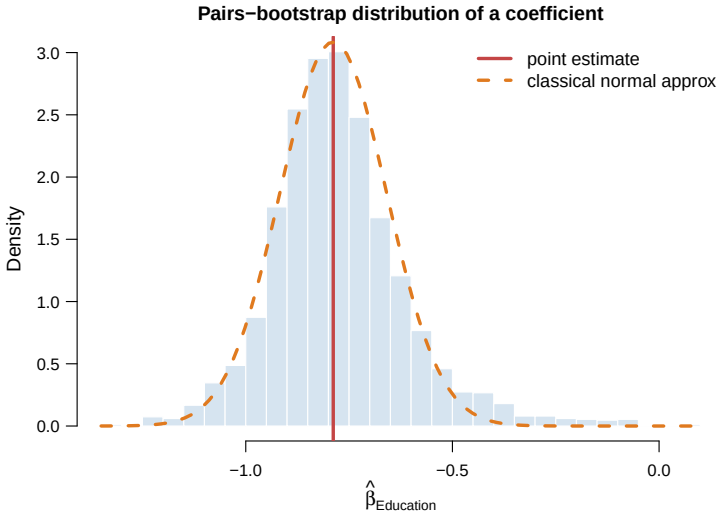
Achieves 93.8% coverage in our simulation — closer to nominal.

Practical comparison

	Heterosk.	Small n
Classical	No	OK
Res. boot	No	OK
Pairs boot	Yes	Fair
HC3	Yes	Good

The bootstrap and HC3 are *complementary*: bootstrap requires no formula; HC3 is faster (no resampling loop).

Bootstrap distribution (Swiss)



Two different questions at x_0

Question A: CI for the mean

What is $\mu_0 = x_0^\top \beta$?

Uncertainty source: *estimation* of β.

$$\hat{\mu}_0 \pm t_{n-p,\alpha/2}\, \hat{\sigma} \sqrt{x_0^\top (X^\top X)^{-1} x_0}$$

Question B: PI for a new observation

What will $y_{\text{new}} = x_0^\top \beta + \varepsilon_{\text{new}}$ be?

Uncertainty sources: estimation *and* irreducible noise.

$$\hat{\mu}_0 \pm t_{n-p,\alpha/2}\, \hat{\sigma} \sqrt{1 + x_0^\top (X^\top X)^{-1} x_0}$$

The $+1$ accounts for $\mathrm{Var}[\varepsilon_{\text{new}}] = \sigma^2$.

As $n \to \infty$: CI $\to$ zero width; PI $\to \pm z_{\alpha/2}\sigma$ (irreducible).

Deriving the "+1": full algebra

Let $\hat{y}_0 = \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}$. For a new draw $y_0 = \mathbf{x}_0^\top \boldsymbol{\beta} + \varepsilon_0$:

$$y_0 - \hat{y}_0 = \underbrace{(\varepsilon_0)}_{\text{new noise}} - \underbrace{\mathbf{x}_0^\top(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})}_{\text{estimation error}}$$

These two components are *independent* (new observation, independent of training data), so:

$$\begin{aligned}
\mathrm{Var}[y_0 - \hat{y}_0] &= \mathrm{Var}[\varepsilon_0] + \mathrm{Var}[\mathbf{x}_0^\top(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})] \\
&= \sigma^2 + \sigma^2\, \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0 \\
&= \sigma^2 \Big(1 + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0\Big)
\end{aligned}$$

Plugging in $\hat{\sigma}^2$ and using t_{n-p} gives the PI formula exactly.

Key takeaway

The CI width $\to 0$ as $n \to \infty$ because estimation improves. The PI width $\to 2z_{\alpha/2}\sigma$ because *individual* noise never averages away.

Numerical example: Swiss Fertility, by hand

Model: `Fertility` $\sim$ `Education`. $n = 47$, $p = 2$ (intercept + slope).

From OLS:

$$\hat{\beta}_0 = 79.61, \quad \hat{\beta}_1 = -0.862, \quad \hat{\sigma} = 10.07$$

$$\bar{x}_{\text{Educ}} = 10.98, \qquad S_{xx} = \sum (x_i - \bar{x})^2 = 2\,826.7$$

Query point: $x_0 = 15$ (Education = 15 years).

$$\hat{\mu}_0 = 79.61 - 0.862 \times 15 = 79.61 - 12.93 = 66.68$$

Leverage of x_0:

$$\mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0 = \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} = \frac{1}{47} + \frac{(15 - 10.98)^2}{2826.7} = 0.02128 + 0.00571 = 0.02699$$

$t_{45,0.025} = 2.014$.

Numerical example: CI and PI computed

95% confidence interval for the mean at $x_0 = 15$:

$$\text{half-width}_{\text{CI}} = 2.014 \times 10.07 \times \sqrt{0.02699} = 2.014 \times 10.07 \times 0.1643 = 3.33$$

$$\boxed{\text{CI}_{0.95}: \quad 66.68 \pm 3.33 \quad = (63.35,\ 70.01)}$$

95% prediction interval for a new observation at $x_0 = 15$:

$$\text{half-width}_{\text{PI}} = 2.014 \times 10.07 \times \sqrt{1 + 0.02699} = 2.014 \times 10.07 \times 1.0134 = 20.54$$

$$\boxed{\text{PI}_{0.95}: \quad 66.68 \pm 20.54 \quad = (46.14,\ 87.22)}$$

The PI is $\approx 6\times$ wider than the CI at this point

Ratio $= 20.54 \,/\, 3.33 = 6.17$. The dominant term is $\hat{\sigma} = 10.07$ — individual noise.

CI and PI in R (Swiss data)

```r
fit0 <- lm(Fertility ~ Education, data = swiss)
x0   <- data.frame(Education = 15)

# 95% confidence interval for the mean response
predict(fit0, newdata = x0, interval = "confidence", level = 0.95)
#       fit       lwr       upr
#    66.683    63.348    70.018

# 95% prediction interval for a new observation
predict(fit0, newdata = x0, interval = "prediction", level = 0.95)
#       fit       lwr       upr
#    66.683    46.132    87.234

# Check: half-widths
(70.018 - 63.348) / 2    # CI half-width: 3.335
(87.234 - 46.132) / 2    # PI half-width: 20.551

# Match the by-hand calculation (minor rounding)
```

The numbers agree with our hand calculation to within rounding.

Width as a function of x_0: the flare

Both CI and PI half-widths are:

$$w_{\mathrm{CI/PI}}(x_0) = t\,\hat{\sigma}\,\sqrt{c + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$$

where $c = 1/n$ (CI) or $c = 1 + 1/n$ (PI).

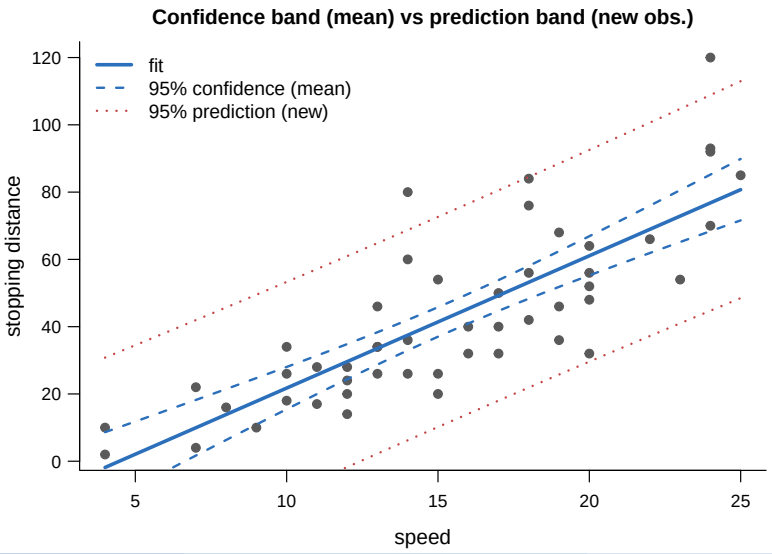
Width is **minimised at** $x_0 = \bar{x}$ (data centroid) and grows as $(x_0 - \bar{x})^2 \nearrow$.

Swiss: half-widths at three query points

x_0	$(x_0 - \bar{x})^2$	CI hw	PI hw
5	35.9	4.71	20.64
11	0.0	2.96	20.42
15	16.2	3.33	20.54
25	198.7	7.41	21.75
40	842.7	14.62	25.01

CI hw doubles between centre and $x_0 = 40$; PI hw increases only modestly because $\hat{\sigma}$ dominates.

Confidence band vs prediction band



Extrapolation: quantifying the danger

Suppose x ranges from 0 to 20 in the training data, so $\bar{x} = 10$ and $S_{xx} = n \cdot \mathrm{Var}(x) \approx 50n/3$ (uniform spacing).

At $x_0 = 30$ (10 units *outside* the range):

$$\mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0 = \frac{1}{n} + \frac{(30-10)^2}{50n/3} = \frac{1}{n} + \frac{400 \cdot 3}{50n} = \frac{1}{n} + \frac{24}{n} = \frac{25}{n}$$

At $x_0 = 10$ (centre): $\mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0 = 1/n$.

Ratio of CI half-widths: extrapolation vs centroid

$$\frac{w_{\mathrm{CI}}(30)}{w_{\mathrm{CI}}(10)} = \frac{\sqrt{25/n}}{\sqrt{1/n}} = 5$$

The CI is **5 times wider** at $x_0 = 30$ than at the centroid. For $n = 50$: centroid CI hw $= 2.0\hat{\sigma}/\sqrt{50}$; extrapolation hw $= 10.0\hat{\sigma}/\sqrt{50}$.

Bootstrap prediction interval (Monte Carlo)

The parametric PI assumes $\varepsilon_{\text{new}} \sim \mathcal{N}(0, \sigma^2)$. A bootstrap PI relaxes this:

```r
fit0  <- lm(Fertility ~ Education, data = swiss)
B     <- 5000;  x0 <- data.frame(Education = 15)
set.seed(7)

boot_pred <- numeric(B)
for (b in seq_len(B)) {
  # Step 1: pairs resample to get beta*
  idx  <- sample(nrow(swiss), replace = TRUE)
  fitb <- lm(Fertility ~ Education, data = swiss[idx, ])
  # Step 2: draw a new epsilon from empirical residual distribution
  eps  <- sample(residuals(fit0), 1)
  # Step 3: predicted new observation
  boot_pred[b] <- predict(fitb, newdata = x0) + eps
}
quantile(boot_pred, c(0.025, 0.975))
#  2.5%    97.5%
#  46.22   87.08          (parametric: 46.13, 87.23)
```

Bootstrap and parametric PIs are nearly identical — consistent with approximate normality of Swiss residuals.

Summary

Bootstrap

- Estimates the sampling distribution by resampling.
- **Pairs**: resample rows; valid under heteroskedasticity.
- **Residual**: resample errors; requires homoskedasticity.
- At small n, under-covers slightly; complemented by HC3.

CI vs PI

- **CI**: uncertainty about $\mu_0 = \mathbf{x}_0^\top \boldsymbol{\beta}$.
- **PI**: uncertainty about a *new draw* y_0.
- PI = CI + σ^2 term; always wider.
- As $n \to \infty$, CI $\to 0$ but PI $\to 2z\sigma$.
- Width is minimised at the data centroid and grows quadratically outside it.

Next

Lecture 6: checking the assumptions — residual diagnostics, outlier detection, and HC3 robust standard errors.

Residual bootstrap in R

```r
fit   <- lm(Fertility ~ Education + Catholic, data = swiss)
yhat  <- fitted(fit);  ehat <- residuals(fit)
B     <- 2000;  set.seed(99)
res_coef <- matrix(NA, B, 3)
for (b in seq_len(B)) {
  e_star         <- sample(ehat, replace = TRUE)
  y_star         <- yhat + e_star
  res_coef[b,]   <- coef(lm(y_star ~ Education + Catholic,
                            data = swiss))
}
se_res <- apply(res_coef, 2, sd)
# Compare: classical SE, pairs SE, residual SE
cbind(classical = round(summary(fit)$coef[,"Std. Error"], 4),
      pairs     = round(apply(boot_coef, 2, sd), 4),    # from earlier
      residual  = round(se_res, 4))
#             classical  pairs  residual
# (Intercept)     7.938  7.821     7.935
# Education       0.392  0.395     0.390
# Catholic        0.073  0.071     0.073
```

All three agree on Swiss data (approximately homoskedastic).

Verify coverage claim by simulation

```r
set.seed(42);  n <- 50;  R <- 3000;  B <- 400
hit <- c(cls=0, res=0, pair=0)
for (r in seq_len(R)) {
  x <- seq(1, 5, length.out = n)
  e <- rnorm(n, sd = x^2)        # heteroskedastic
  y <- 1 + 2*x + e;  dat <- data.frame(x, y)
  fit <- lm(y ~ x, data = dat)
  ci  <- confint(fit)["x",]
  hit["cls"] <- hit["cls"] + (ci[1] < 2 & 2 < ci[2])
  yh <- fitted(fit);  eh <- residuals(fit)
  bc <- pc <- numeric(B)
  for (b in seq_len(B)) {
    idx    <- sample(n, replace = TRUE)
    bc[b]  <- coef(lm(y ~ x, data = dat[idx,]))["x"]
    pc[b]  <- coef(lm(yh + sample(eh,n,replace=TRUE) ~ x,
                      data = dat))["x"]
  }
  ci_p  <- quantile(bc, c(.025,.975))
  ci_r  <- quantile(pc, c(.025,.975))
  hit["pair"] <- hit["pair"] + (ci_p[1]<2 & 2<ci_p[2])
  hit["res"]  <- hit["res"]  + (ci_r[1]<2 & 2<ci_r[2])
}
```